

Back-off Models

- **The idea:** assume we want to estimate a trigram distribution for a word x_m given its two previous words x_{m-1} and x_{m-2} :

$$q_{ML}(x_m = w_j | x_{m-1} = w_i, x_{m-2} = w_l) = \frac{\text{count}(w_l, w_i, w_j)}{\sum_{j'} \text{count}(w_l, w_i, w_{j'})}$$

- If the bigram (x_{m-1}, x_{m-2}) didn't appear enough times, we back-off to conditioning only on x_{m-1} :

$$q_{ML}(x_m = w_j | x_{m-1} = w_i) = \frac{\text{count}(w_i, w_j)}{\sum_{j'} \text{count}(w_i, w_{j'})}$$

- If w_{i-1} is not frequent enough, we back-off to w_i 's frequency:

$$q_{ML}(x_m = w_j) = \frac{\text{count}(w_j)}{\sum_{j'} \text{count}(w_{j'})}$$

Notation

- For brevity, we will assume that the transitions are not dependent on m (“homogeneous Markov models”) and use the notation

$$P(w_j | w_i, w_l)$$

instead of

$$P(x_m = w_j | x_{m-1} = w_i, x_{m-2} = w_l)$$